

**NORWICH INTL, United Kingdom**WMO: **034920**Lat: **52.676N**Lon: **1.283E**Elev: **117**StdP: **14.63**Time Zone: **0.00 (EUW)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 28.0       | 30.0 | 21.3                          | 16.1 | 32.6 | 24.1 | 18.3 | 34.0 | 27.9                  | 48.1 | 25.6 | 48.0 | 7.7                   | 80   | 0.571 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 15.2                   | 79.9            | 65.8 | 76.6 | 64.1 | 73.5 | 62.8 | 67.5                | 76.1 | 65.7 | 73.3 | 64.3 | 71.2 | 9.7                  | 110  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 64.4                            | 91.1 | 71.0 | 62.8 | 85.9 | 68.8 | 61.3 | 81.5 | 67.1 | 31.9          | 76.6 | 30.6 | 73.3 | 29.5 | 71.4 |                   |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 23.1          | 20.1 | 17.8 | DB                         | 22.9 | 87.3               | 4.2 | 3.0                                                | 19.9 | 89.4       | 17.5 | 91.1       | 15.1 | 92.8       | 12.1 | 95.0 | (4) |
| (5)               |      |      | WB                         | 22.1 | 71.3               | 4.0 | 1.6                                                | 19.3 | 72.5       | 16.9 | 73.4       | 14.7 | 74.3       | 11.8 | 75.5 | (5) |

**Monthly Climatic Design Conditions**

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 51.2        | 40.3  | 41.6  | 44.5  | 47.9  | 53.5  | 58.9  | 63.7  | 63.7  | 58.4  | 52.7  | 45.8  | 42.3  | (6)  |  |
|                                                                        | DBStd       | 9.70        | 5.48  | 5.53  | 5.37  | 5.47  | 5.58  | 4.78  | 4.68  | 4.68  | 4.49  | 5.01  | 5.37  | 5.80  | (7)  |  |
|                                                                        | HDD50       | 1276        | 301   | 239   | 182   | 104   | 27    | 0     | 0     | 0     | 1     | 29    | 148   | 245   | (8)  |  |
|                                                                        | HDD65       | 5146        | 764   | 656   | 635   | 513   | 357   | 192   | 81    | 83    | 205   | 382   | 576   | 702   | (9)  |  |
|                                                                        | CDD50       | 1704        | 1     | 3     | 13    | 41    | 136   | 267   | 424   | 425   | 252   | 112   | 22    | 8     | (10) |  |
|                                                                        | CDD65       | 100         | 0     | 0     | 0     | 0     | 1     | 9     | 40    | 43    | 6     | 1     | 0     | 0     | (11) |  |
|                                                                        | CDH74       | 591         | 0     | 0     | 0     | 2     | 11    | 65    | 255   | 225   | 32    | 1     | 0     | 0     | (12) |  |
|                                                                        | CDH80       | 98          | 0     | 0     | 0     | 0     | 0     | 12    | 46    | 36    | 4     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 8.5         | 9.6   | 9.6   | 9.3   | 8.4   | 8.2   | 7.9   | 7.4   | 7.7   | 7.7   | 8.2   | 8.3   | 9.8   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 25.5        | 2.2   | 1.6   | 1.9   | 1.9   | 1.8   | 2.0   | 2.1   | 2.3   | 2.1   | 2.4   | 2.7   | 2.5   | (15) |  |
|                                                                        | PrecMax     | 32.8        | 3.9   | 4.0   | 4.0   | 4.5   | 3.8   | 4.6   | 4.2   | 4.4   | 6.5   | 5.2   | 5.7   | 4.9   | (16) |  |
|                                                                        | PrecMin     | 20.3        | 0.5   | 0.2   | 0.5   | 0.2   | 0.4   | 0.4   | 0.3   | 0.3   | 0.3   | 0.4   | 1.1   | 1.1   | (17) |  |
|                                                                        | PrecStd     | 3.4         | 0.9   | 0.9   | 0.9   | 0.9   | 0.9   | 1.2   | 1.1   | 1.1   | 1.4   | 1.5   | 1.0   | 0.9   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 55.7  | 59.1  | 62.5  | 70.0  | 75.4  | 80.8  | 85.7  | 84.4  | 78.3  | 69.5  | 61.0  | 57.6  | (19) |  |
|                                                                        |             | MCWB        | 52.0  | 52.2  | 53.6  | 57.5  | 61.5  | 65.4  | 68.1  | 65.9  | 65.8  | 61.9  | 58.1  | 54.2  | (20) |  |
|                                                                        | 2%          | DB          | 53.2  | 54.1  | 58.6  | 64.0  | 71.1  | 75.0  | 80.2  | 79.6  | 73.0  | 65.2  | 58.8  | 55.7  | (21) |  |
|                                                                        |             | MCWB        | 50.3  | 49.4  | 51.8  | 54.5  | 60.7  | 63.6  | 66.4  | 65.2  | 63.2  | 59.6  | 55.7  | 52.9  | (22) |  |
|                                                                        | 5%          | DB          | 50.4  | 51.8  | 55.5  | 60.0  | 66.8  | 71.5  | 76.8  | 75.8  | 68.4  | 62.6  | 56.0  | 53.7  | (23) |  |
|                                                                        |             | MCWB        | 47.9  | 48.0  | 50.3  | 52.1  | 57.5  | 61.7  | 64.1  | 63.9  | 60.8  | 58.3  | 53.5  | 51.2  | (24) |  |
|                                                                        | 10%         | DB          | 48.4  | 49.9  | 53.2  | 57.1  | 63.8  | 68.1  | 73.5  | 72.8  | 66.0  | 60.5  | 53.8  | 51.6  | (25) |  |
|                                                                        |             | MCWB        | 46.4  | 46.8  | 48.5  | 50.9  | 56.0  | 59.9  | 63.0  | 62.3  | 59.6  | 56.7  | 51.6  | 49.5  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 53.4  | 53.2  | 54.6  | 59.1  | 64.5  | 68.3  | 70.1  | 69.1  | 68.3  | 63.0  | 58.9  | 55.2  | (27) |  |
|                                                                        |             | MCDB        | 55.4  | 57.3  | 59.8  | 68.0  | 72.4  | 77.6  | 81.8  | 79.6  | 74.7  | 67.9  | 60.7  | 57.2  | (28) |  |
|                                                                        | 2%          | WB          | 50.6  | 50.4  | 52.6  | 55.5  | 61.4  | 65.3  | 68.0  | 66.9  | 64.6  | 60.8  | 56.5  | 53.6  | (29) |  |
|                                                                        |             | MCDB        | 52.3  | 53.4  | 57.2  | 61.6  | 68.6  | 72.4  | 77.6  | 75.3  | 70.5  | 64.1  | 58.4  | 55.9  | (30) |  |
|                                                                        | 5%          | WB          | 48.4  | 48.5  | 50.9  | 53.3  | 59.0  | 63.0  | 65.8  | 65.3  | 62.6  | 59.0  | 54.1  | 51.8  | (31) |  |
|                                                                        |             | MCDB        | 50.0  | 51.1  | 54.9  | 58.9  | 65.6  | 68.9  | 73.3  | 73.2  | 66.8  | 61.7  | 55.8  | 53.7  | (32) |  |
|                                                                        | 10%         | WB          | 46.4  | 46.9  | 49.1  | 51.4  | 56.7  | 61.0  | 64.2  | 63.8  | 60.8  | 57.2  | 52.2  | 49.7  | (33) |  |
|                                                                        |             | MCDB        | 48.1  | 49.5  | 52.7  | 56.2  | 62.6  | 67.0  | 71.0  | 70.8  | 64.6  | 60.1  | 53.9  | 51.4  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 8.7   | 9.9   | 11.4  | 13.0  | 14.4  | 14.1  | 15.2  | 15.3  | 13.5  | 11.7  | 9.8   | 8.5   | (35) |  |
|                                                                        |             | MCDBR       | 11.1  | 11.8  | 15.3  | 17.6  | 20.2  | 20.0  | 21.4  | 19.7  | 17.2  | 13.7  | 9.9   | 10.1  | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 10.6  | 9.2   | 10.5  | 10.2  | 11.3  | 11.0  | 9.3   | 8.4   | 9.4   | 8.6   | 8.6   | 9.4   | (37) |  |
|                                                                        |             | MCWBR       | 10.9  | 11.7  | 14.9  | 16.5  | 19.0  | 17.8  | 18.6  | 17.6  | 15.7  | 12.8  | 9.7   | 9.8   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.326 | 0.343 | 0.377 | 0.400 | 0.403 | 0.407 | 0.408 | 0.397 | 0.377 | 0.356 | 0.330 | 0.315 | (40) |  |
|                                                                        |             | taud        | 2.349 | 2.372 | 2.277 | 2.236 | 2.281 | 2.286 | 2.297 | 2.344 | 2.398 | 2.426 | 2.397 | 2.336 | (41) |  |
|                                                                        | Ebn at Noon | Ebn at Noon | 208   | 242   | 255   | 264   | 269   | 268   | 266   | 262   | 253   | 234   | 205   | 188   | (42) |  |
|                                                                        |             | Edh at Noon | 19    | 25    | 33    | 39    | 39    | 40    | 39    | 35    | 30    | 23    | 18    | 16    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 225   | 442   | 802   | 1262  | 1544  | 1627  | 1558  | 1316  | 979   | 566   | 281   | 179   | (44) |  |
|                                                                        | RadStd      | RadStd      | 22    | 65    | 85    | 130   | 117   | 123   | 157   | 92    | 81    | 49    | 29    | 21    | (45) |  |

**Historical Trends**

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (66 neighbors) | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page